OS W7 Practical Assignment

Inode Simulation — Sector Cost Calculation & Addressing Method

1. Bash Script (inode_sim.sh)

```bash
#!/bin/bash

# Inode Simulation - Sector Cost Calculation

SECTOR_SIZE=512
DIRECT_POINTERS=12
MAX_DIRECT_SIZE=$((SECTOR_SIZE * DIRECT_POINTERS))

# File sizes in bytes (binary units)
declare -a FILE_SIZES=("1 KB" "100 KB" "1 MB" "1 GB" "10 GB")
declare -a BYTE_COUNTS=(
    $((1 * 1024))
    $((100 * 1024))
    $((1 * 1024 * 1024))
    $((1 * 1024 * 1024 * 1024))
    $((10 * 1024 * 1024 * 1024))
)

# Print the table header
printf "%-12s %-15s %-15s %s\n" \
    "File Size" "Byte Count" "Sector Count" "Addressing Method"

printf '%s\n' "--------------------------------------------------------------------------------"

# Calculate sector count and addressing method
for i in "${!FILE_SIZES[@]}"; do
    size="${FILE_SIZES[$i]}"
    bytes="${BYTE_COUNTS[$i]}"

    # Round up to the next whole sector
    sectors=$(((bytes + SECTOR_SIZE - 1) / SECTOR_SIZE))

    if (( bytes > MAX_DIRECT_SIZE )); then
        method="INDIRECT ADDRESSING NEEDED"
    else
        method="DIRECT ADDRESSING SUFFICIENT"
    fi

    printf "%-12s %-15s %-15s %s\n" \
        "$size" "$bytes" "$sectors" "$method"
done
```

2. Terminal Execution and Output

```
rohit01DELL13WINDOW11:~$ ls
OS_Week6   ROS    h:txt   hello.txt  hello2.txt  ignorant  ls.txt  rr.sh
rohit01DELL13WINDOW11:~$ nano inode_sim.sh
rohit01DELL13WINDOW11:~$ chmod +x inode_sim.sh
rohit01DELL13WINDOW11:~$ ./inode_sim.sh
File Size     Byte Count     Sector Count    Addressing Method
--------------------------------------------------------------------------
1 KB          1024           2               DIRECT ADDRESSING SUFFICIENT
100 KB        102400         200             INDIRECT ADDRESSING NEEDED
1 MB          1048576        2048            INDIRECT ADDRESSING NEEDED
1 GB          1073741824     2097152         INDIRECT ADDRESSING NEEDED
10 GB         10737418240    20971520        INDIRECT ADDRESSING NEEDED
rohit01DELL13WINDOW11:~$
```